

Omer Attia

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[Github](#) | [Linkedin](#) | [Website](#)

Software Engineering graduate (August 2026) specializing in Machine Learning and Embedded Systems, with strong foundations in data structures, algorithms, databases, and systems programming. I have hands-on experience with Linux-based systems, networking, Docker, and embedded hardware, complemented by personal projects in self-hosting and electronics. I'm looking to bring that same hands-on mindset into a professional setting.

EDUCATION

Afeka College Of Engineering

2022 – 2026

Software Engineering

Specializing in ML & Embedded systems, progressing through a comprehensive degree with strong performance in core subjects, including data structures, algorithms, databases, networks, embedded programming and systems programming.

Kernelios

2022 – 2023

Cyber Security Implementer

Built a strong foundation in cybersecurity, with practical knowledge in network security, domain management (Windows Server 2022), and key security tools and techniques.

WORK EXPERIENCE

Ichilov Hospital

Customer Service Representative

July 2022 – July 2024

Demonstrated excellent communication skills to effectively interact with patients. Provided administrative services to the patients and their families.

MILITARY EXPERIENCE

IDF

Squad Commander

March 2019 – December 2021

Served as a commander in the ground defense division of the Air Force. As part of my role I performed emergency drills, was in charge of staff training and the implementation of security principles in my unit.

SKILLS

Programming: C, C++, Python, Java, Rust

Domain Expertise: Embedded, Data Structures & Algorithms, ML, CV, Concurrency, Compilation, OS, Networks

Tools & Frameworks: Git, Linux, Docker, SQL, Flutter, Supabase, Spring, STM, RPi, ESP32, FreeRTOS

Attributes: Problem-solving, Leadership, Teamwork, Training & Mentoring

Languages: Hebrew (Native) · English (Advanced)

CERTIFICATIONS

Linux Essentials | Linux Professional Institute

Windows Domain Management | Kernelios

Certified Hands-on Cyber Security Specialist (CHCSS) | Kernelios

AWARDS

Afekathon 2026: 1st Place · Afeka College Hackathon

Campus Navigation System - Indoor/outdoor navigation using GPS outdoors and WiFi fingerprinting indoors (MAC address + RSSI sampling mapped to campus coordinates).

PROJECTS

GDO Compiler: Designed and implemented a compiler in Python for a functional, lambda-oriented language - including lexing, parsing, and evaluation stages.

Basys MX3 Microcontroller Game: Developed game for PIC32-based microcontroller, working directly with hardware peripherals (switches, buttons, LCD display, etc.) and memory-mapped I/O with no OS or HAL.

Entrance System: End-to-end RFID access control system with Arduino firmware (C++), an Axum (Rust) backend with SQLite logging and Server-Sent Events, and a Flutter dashboard for real-time access monitoring.

Accessible Traffic Light (*Afeka's Engineers Without Borders*): Traffic light commissioned by a special-education school. ESP32 firmware, FreeRTOS, GPIO interrupt with task notification. Self-hosted WiFi AP with captive portal for browser control. Designed and soldered the power/control chain (buck converter, dual relay, audio module and a physical button).

Bina System (Final Project): A dental diagnosis system, Flutter app with YOLO-based intra-oral detection, dual backend (Supabase + SQLite), integrated on-device LLM and Embeddings for medical history queries. Trained CV model on tagged dataset, 3D-modeled camera case, developed RPi firmware for video streaming and peripheral control.

[For more projects...](#)